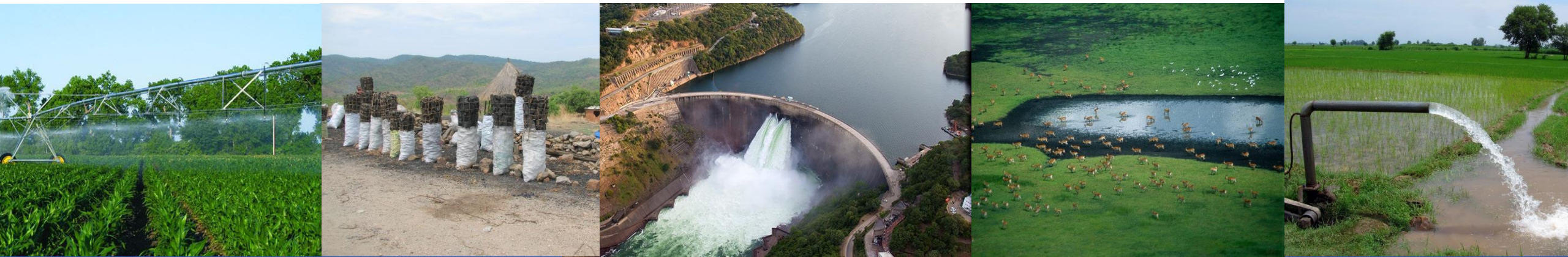


Preparation: Hands on CWatM

Peter Burek, Carla Catania, Silvia Artuso
International Institute for Applied Systems Analysis
Research Scholars at
Water Research Group, BNR

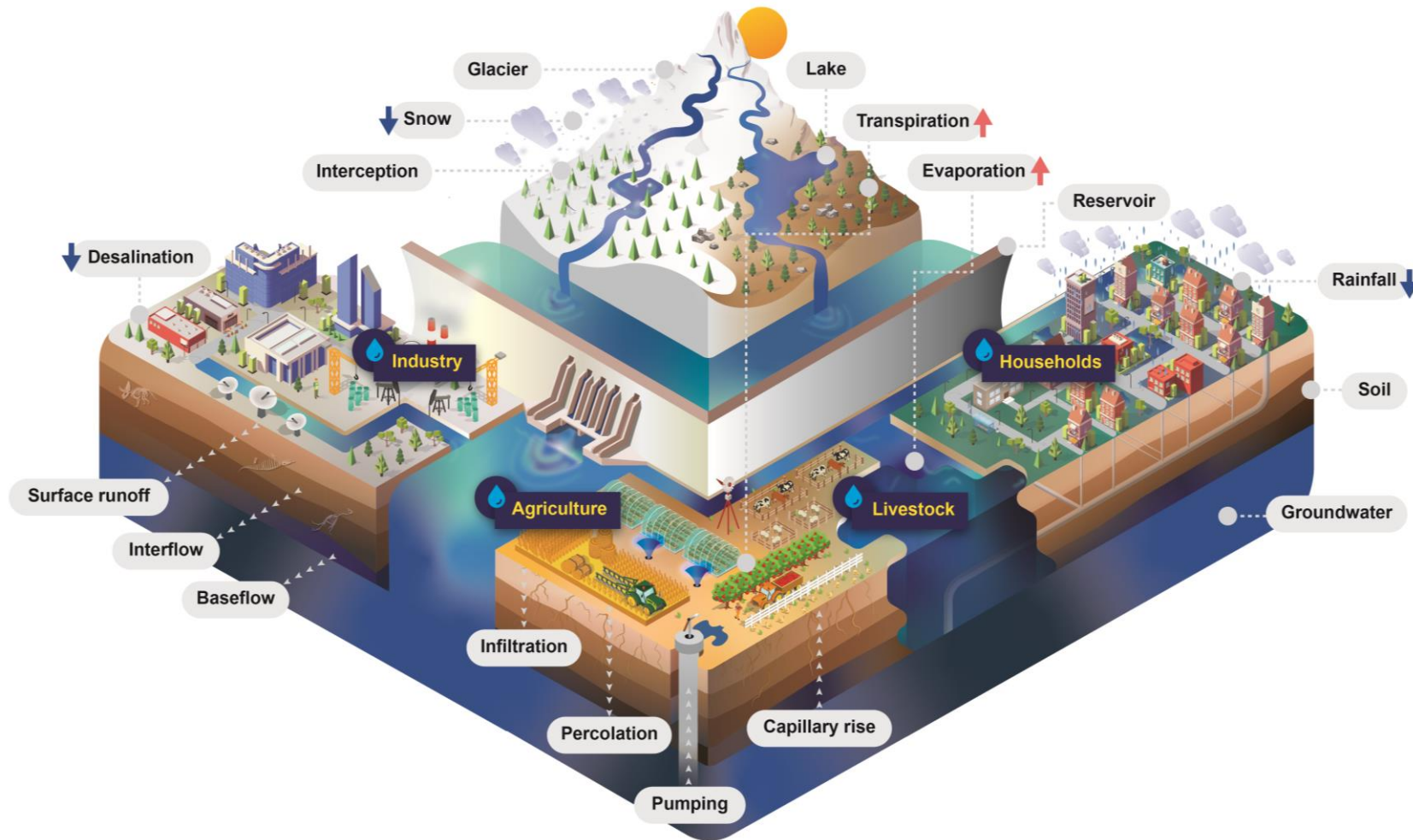


Hands on CWatM

1. Downloading data
2. Installing software
3. Installing Python, libraries and CWatM
4. Running CWatM for the first time
5. Run CWatM with a settings file
6. Have a look at the documentation <https://cwatm.iiasa.ac.at/>
7. Have a look at github <https://github.com/iiasa/CWatM>



Community Water Model (CWatM)



The hydrological model CWatM

- can be used globally ($0.5^\circ, 5'$) and regionally ($5', 1'$)
- is open source in python
- it used globally freely available data
- it includes the major hydrological processes

<http://www.iiasa.ac.at/cwatm>

<https://cwatm.iiasa.ac.at/>

Training workshop – Phase 0

Preparation

IIASA

- Documentation and weblinks: <https://cwatm.iiasa.ac.at/>
- CWatM manual on: <http://pure.iiasa.ac.at/id/eprint/16802/>
- Main literature on SharePoint: [SO2/working/CWatM_model_developversion](https://iiasahub.sharepoint.com/:f:/r/teams/DanubeWaterBalance/SO2/working/CWatM_model_developversion)
- You Tube videos: <https://t.ly/TvDny>
- SharePoint server:

<https://iiasahub.sharepoint.com/:f:/r/teams/DanubeWaterBalance>

Specific the SO2/working folder

<https://iiasahub.sharepoint.com/:f:/r/teams/DanubeWaterBalance/Shared%20Documents/General/SO2/working?csf=1&web=1&e=zXpfie>

Participants (till 10th December):

- Installation of a texteditor and netCDF viewer, Python 3.x and libraries if you did not do it yet
- Installation of Python and CWatM
- Download of data and model from SharePoint

CWatM tutorial series to run CWatM

- Setting up Python and GitHub
- Running CWatM
- Comparing with observations of changes in water storage, using Google Earth Engine (GEE)
- Calibrating with a river discharge station

Learning the hydrological model CWatM with Silvia, 1: Setting up CWatM
Mikhail Smilovic • 76 Aufrufe • vor 2 Monaten

How to install Python, GDAL, virtualenvwrapper & other packages on Windows
Mikhail Smilovic • 158 Aufrufe • vor 4 Monaten

Learning the hydrological model CWatM with Silvia, 2: Beginning the Danube basin
Mikhail Smilovic • 67 Aufrufe • vor 1 Monat

Creating initialisation files: CWatM with Silvia, 3
Mikhail Smilovic • 16 Aufrufe • vor 13 Tagen

Creating Water circles: Hydrological model CWatM with Silvia, 4
Mikhail Smilovic • 11 Aufrufe • vor 12 Tagen

Coding in CWatM and pull requests: Hydrological model CWatM with Silvia, 5
Mikhail Smilovic • 17 Aufrufe • vor 11 Tagen

Using GRACE with Google Earth Engine for changes in water storage over the Danube basin
Mikhail Smilovic • 56 Aufrufe • vor 6 Tagen

Comparing GRACE and CWatM for the Danube basin, and observing changes in water storage
Mikhail Smilovic • 28 Aufrufe • vor 3 Tagen

Documentation: <https://cwatm.iiasa.ac.at>
Tutorials: <https://t.ly/TvDny>

Installation of software 1

Texteditor

CWatM information (settingsfile) is stored in a text file, which can be read by Windows notepad.

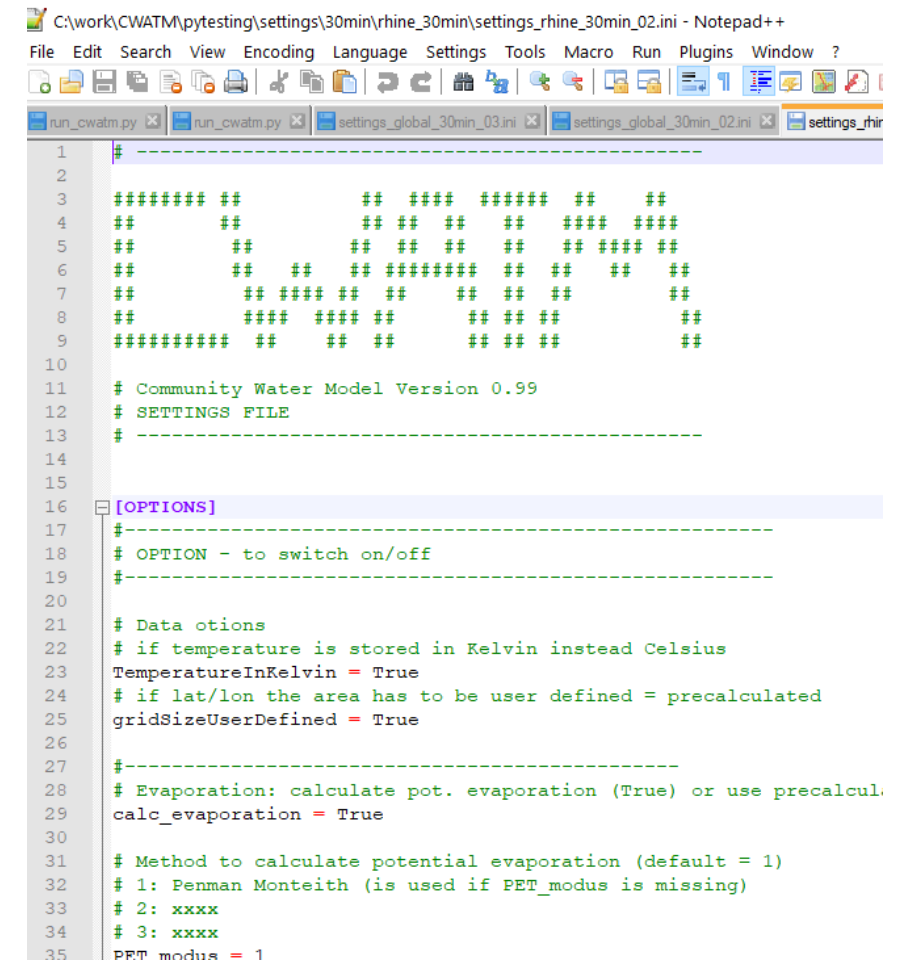
But we strongly recommend a better text editor!

You can use any text editor you are familiar with.

In case you do not have a texteditor

Install notepad++ from

<https://notepad-plus-plus.org>



```
1  # -----
2
3  #####  ##          ##  ###  #####  ##  ##
4  ##      ##          ##  ##  ##  #####  ##
5  ##      ##          ##  ##  ##  ##  ##  ##
6  ##      ##  ##  ##  #####  ##  ##  ##  ##
7  ##      ##  ##  ##  ##  ##  ##  ##  ##
8  ##      ###  ###  ##  ##  ##  ##  ##
9  #####  ##  ##  ##  ##  ##  ##
10
11  # Community Water Model Version 0.99
12  # SETTINGS FILE
13  # -----
14
15
16  [OPTIONS]
17  # -----
18  # OPTION - to switch on/off
19  # -----
20
21  # Data options
22  # if temperature is stored in Kelvin instead Celsius
23  TemperatureInKelvin = True
24  # if lat/lon the area has to be user defined = precalculated
25  gridSizeUserDefined = True
26
27  #-----
28  # Evaporation: calculate pot. evaporation (True) or use precalcul
29  calc_evaporation = True
30
31  # Method to calculate potential evaporation (default = 1)
32  # 1: Penman Monteith (is used if PET_modus is missing)
33  # 2: xxxx
34  # 3: xxxx
35  PET_modus = 1
```

Installation of software 2

NetCDF viewer

CWatM data and output can be quite large

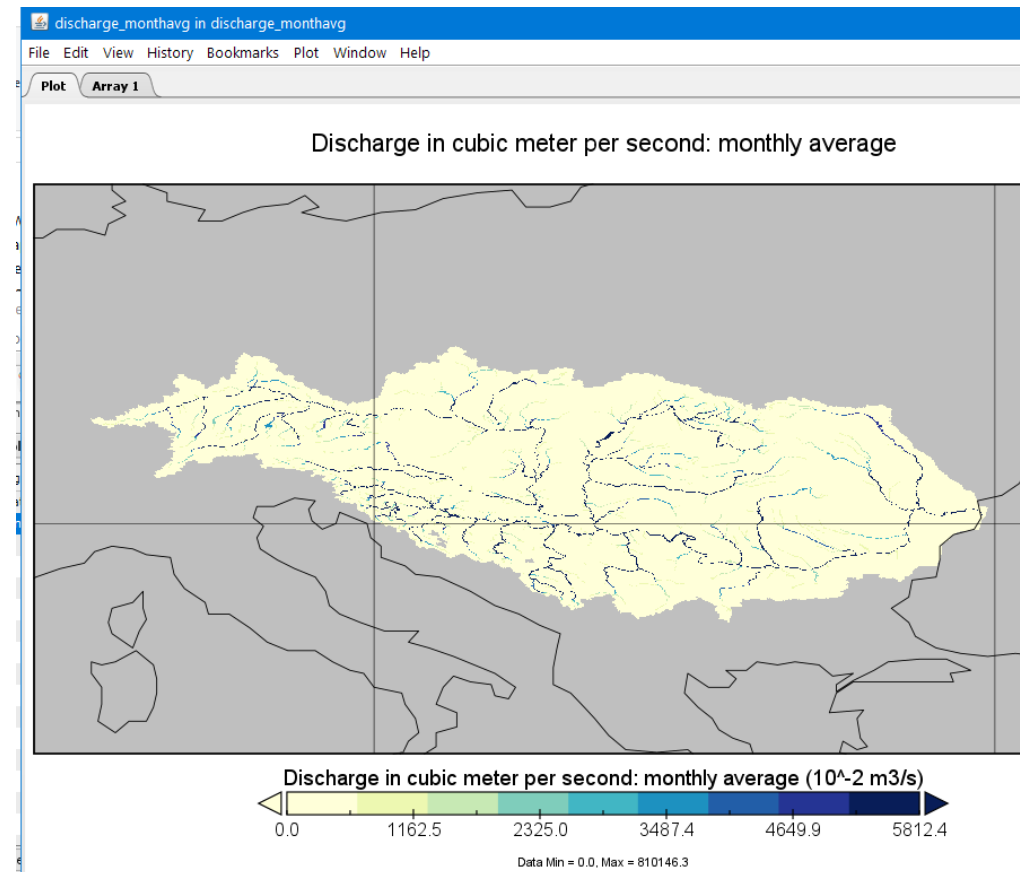
Therefore we store our data as compressed as possible as netCDF files

NetCDF is a nice format but not so easy to handle as this is mostly used by meteorologist on there Linux systems

To view input and output data on Windows you can use the Panoply netCDF viewer

Install panoply from the Dropbox from:

<https://www.giss.nasa.gov/tools/panoply/>



Installation of software 3a

CWatM Python

Our software is mainly programmed as Python3.x source code

- One of the easiest programming language
- The source code is freely available: <https://cwatm.iiasa.ac.at/sourcecode.html>
- The way to contribute to CWatM is to get familiar with Python

Here we give NO Python course, as CWatM also runs without any programming skills!

But we strongly recommend to install Python and the necessary libraries to be up to date with our newest versions on GitHub

Martin Bednář wrote a really nice setup document:

SO2/working/CWatM_model_developv
ersion/CWatM_setup.pdf

Please have a look at the YouTube video: <https://www.youtube.com/watch?v=gW6AUELHvu0>

Installation of software 3b

CWatM Python

Python 3.12 64 bit you can find:

On our Sharepoint: SO2/working/CWatM_model_developversion/python_gdal_library

Or on:

<https://www.python.org/downloads/release/python-386/>

You also can use Anaconda Python.

To run CWatM you need also some libraries

We have limited the use of libraries to a minimum but those one you need:

- numpy
- netCDF4
- xmipy
- GDAL
- scipy
- pandas
- openpyxl

And for the calibration tool: deap and matplotlib

Please read: <https://cwatm.iiasa.ac.at/setup.htm>

Installation of software 3c

CWatM Python libraries

Please read: <https://cwatm.iiasa.ac.at/setup.htm>

But in most cases you can install them with pip (or conda if you use anaconda)

pip install numpy

pip install scipy

pip install netCDF4

pip install pandas

pip install xmipy

pip install openpyxl

pip install your_path_to/GDAL-3.9.2-cp312-cp312-win_amd64

And for calibration

pip install deap

pip install matplotlib

and for postprocessing

pip install plotly

pip install Jupyter Notebook

pip install XlsxWriter

Command Prompt

```
Microsoft Windows [Version 10.0.17763.1217]  
(c) 2018 Microsoft Corporation. All rights reserved.
```

```
H:\>c:
```

```
C:\>cd Python37
```

```
C:\Python37>pip install numpy
```

Installation of software 3d

CWatM Python libraries

Troublemaker GDAL library

In most cases

Pip install GDAL will not work!

We provide the GDAL library for Python 3.12 in
SO2/working/CWatM_model_developversion/python_gdal_library

[GDAL-3.9.2-cp312-cp312-win_amd64](#)

If you use another version you can download the library:

<https://github.com/cgohlke/geospatial-wheels/releases>

Please install

```
pip install your_path_to/GDAL-3.9.2-cp312-cp312-win_amd64
```

Please have a look at: <https://cwatm.iiasa.ac.at/setup.html#external-libraries>

Martin Bednář wrote a really nice setup document:

[SO2/working/CWatM_model_developversion/CWatM_setup.pdf](#)

Installation of software 4a

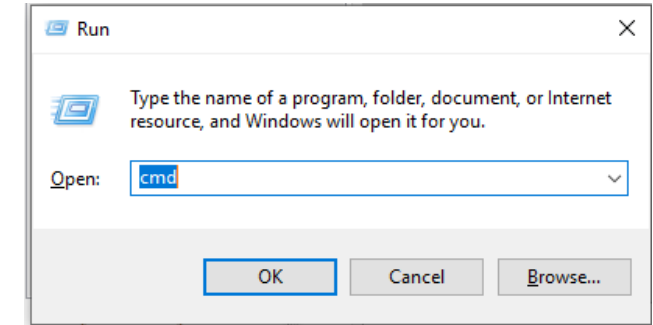
CWatM

Download CWatM from SharePoint:

- [SO2/working/CWatM_model_developversion](#)

Or:

<https://github.com/iiasa/CWatM/tree/develop>



Open a DOS command prompt

- press Windows+X
- type cmd + return
- change directory e.g.: cd C:/CWATM
(or cd "c:/directory with white space/CWATM")

Type python run_cwatm.py

```
P:\watmodel\CWatM\Regions\Danube_1min\CWatM_model_developversion>python run_cwatm.py
CWatM - Community Water Model
Authors: Peter Burek, Mikhail Smilovic, Jens de Bruijn, Dor Fridman, Luca Guillaumot, Yusuke Satoh, Peter
Hanus, Jessica Fennel
Version: 1.5
Date: 01/01/2023
Status: Development

Arguments list:
settings.ini    settings file

-q --quiet      output progression given as .
-v --veryquiet no output progression is given
-l --loud       output progression given as time step, date and discharge
-c --check      input maps and stack maps are checked, output for each input map BUT no model run
-t --printtime  the computation time for hydrological modules are printed
-w --warranty   copyright and warranty information

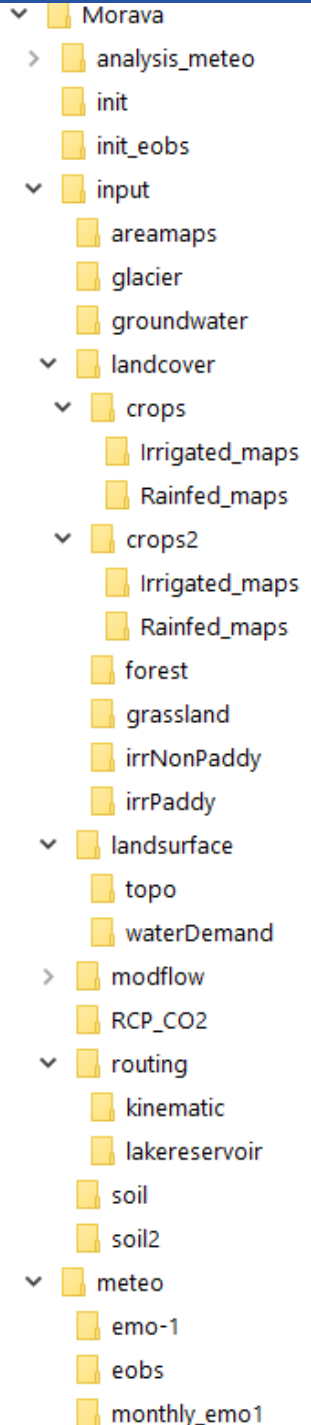
P:\watmodel\CWatM\Regions\Danube_1min\CWatM_model_developversion>
```

Ist running run on Windows, Linux,
For Apple it might run.

If it is not working, please have a look: <https://cwatm.iiasa.ac.at/setup.html#test-the-python-version>

Installation of software 4b

CWatM



Installation of software 4c

CWatM settings file

The setting file for CWatM is in the folder:

Settings:

e.g. `/CWatM_1min/Sava/settings_drina_1min.ini`

It should run with the relative paths given in
[File_Paths]

In case it did not run, please fill in your
absolute paths

As you can see the settingsfile has a
lot of options:

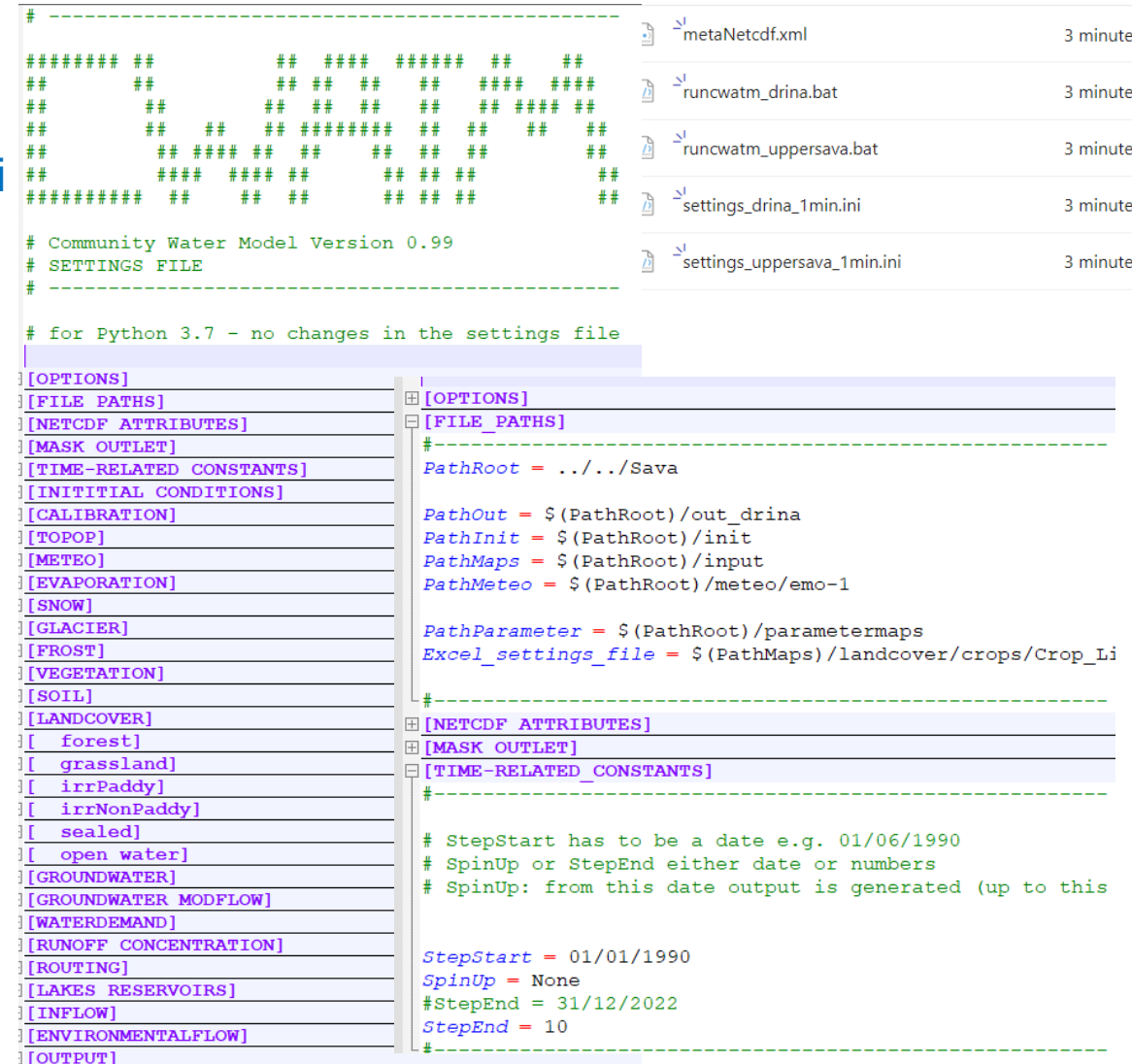
The most important are at the beginning
and at the end:

[Options]

[File_Paths]

[Time_related_constants]

[Output]



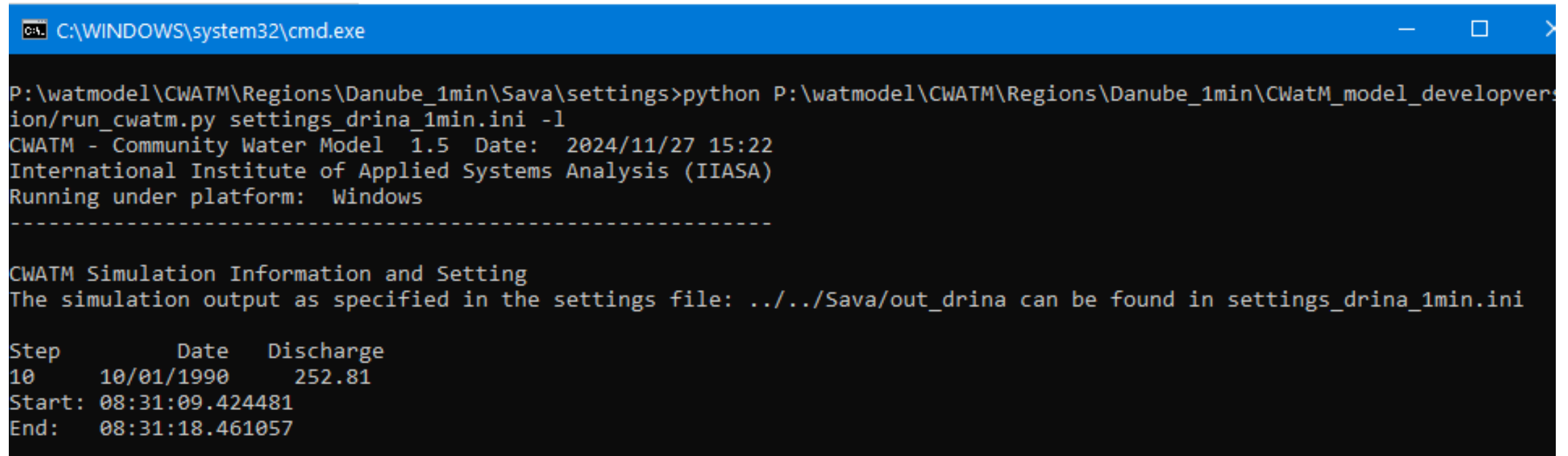
The image shows a file explorer window on the right and a code editor on the left. The file explorer is titled 'General' and shows a directory structure: Documents > General > SO2 > working > CWatM_1min > Sava > settings. It lists several files: metaNetcdf.xml, runcwatm_drina.bat, runcwatm_uppersava.bat, settings_drina_1min.ini, and settings_uppersava_1min.ini, all modified 3 minutes ago. The code editor on the left shows the content of the 'settings_drina_1min.ini' file. It starts with a header section containing a dashed line, a title '#####', and a description 'Community Water Model Version 0.99 SETTINGS FILE'. Below this is a section for Python 3.7. The main body of the file is divided into several sections: [OPTIONS], [FILE_PATHS], [NETCDF ATTRIBUTES], [MASK OUTLET], [TIME-RELATED CONSTANTS], [INITIAL CONDITIONS], [CALIBRATION], [TOPOG], [METEO], [EVAPORATION], [SNOW], [GLACIER], [FROST], [VEGETATION], [SOIL], [LANDCOVER], [forest], [grassland], [irrPaddy], [irrNonPaddy], [sealed], [open water], [GROUNDWATER], [GROUNDWATER MODFLOW], [WATERDEMAND], [RUNOFF CONCENTRATION], [ROUTING], [LAKES RESERVOIRS], [INFLOW], [ENVIRONMENTALFLOW], and [OUTPUT]. The [FILE_PATHS] section contains relative paths for PathRoot, PathOut, PathInit, PathMaps, PathMeteo, PathParameter, and Excel_settings_file. The [TIME-RELATED CONSTANTS] section contains StepStart, SpinUp, StepEnd, and SpinUp values.

```
# -----  
#####  
# Community Water Model Version 0.99  
# SETTINGS FILE  
# -----  
  
# for Python 3.7 - no changes in the settings file  
[OPTIONS]  
[FILE_PATHS]  
[NETCDF ATTRIBUTES]  
[MASK OUTLET]  
[TIME-RELATED CONSTANTS]  
[INITIAL CONDITIONS]  
[CALIBRATION]  
[TOPOG]  
[METEO]  
[EVAPORATION]  
[SNOW]  
[GLACIER]  
[FROST]  
[VEGETATION]  
[SOIL]  
[LANDCOVER]  
[ forest]  
[ grassland]  
[ irrPaddy]  
[ irrNonPaddy]  
[ sealed]  
[ open water]  
[GROUNDWATER]  
[GROUNDWATER MODFLOW]  
[WATERDEMAND]  
[RUNOFF CONCENTRATION]  
[ROUTING]  
[LAKES RESERVOIRS]  
[INFLOW]  
[ENVIRONMENTALFLOW]  
[OUTPUT]  
  
[OPTIONS]  
[FILE_PATHS]  
# -----  
PathRoot = ../../Sava  
  
PathOut = $(PathRoot)/out_drina  
PathInit = $(PathRoot)/init  
PathMaps = $(PathRoot)/input  
PathMeteo = $(PathRoot)/meteo/emo-1  
  
PathParameter = $(PathRoot)/parametermaps  
Excel_settings_file = $(PathMaps)/landcover/crops/Crop_Li  
# -----  
[NETCDF ATTRIBUTES]  
[MASK OUTLET]  
[TIME-RELATED CONSTANTS]  
# -----  
# StepStart has to be a date e.g. 01/06/1990  
# SpinUp or StepEnd either date or numbers  
# SpinUp: from this date output is generated (up to this  
  
StepStart = 01/01/1990  
SpinUp = None  
#StepEnd = 31/12/2022  
StepEnd = 10  
# -----
```

Installation of software 4d

Running CWatM

Open a DOS command prompt



```
C:\WINDOWS\system32\cmd.exe

P:\watmodel\CWATM\Regions\Danube_1min\Sava\settings>python P:\watmodel\CWATM\Regions\Danube_1min\CWatM_model_developversion/run_cwatm.py settings_drina_1min.ini -l
CWATM - Community Water Model 1.5 Date: 2024/11/27 15:22
International Institute of Applied Systems Analysis (IIASA)
Running under platform: Windows
-----

CWATM Simulation Information and Setting
The simulation output as specified in the settings file: ../../Sava/out_drina can be found in settings_drina_1min.ini

Step      Date      Discharge
10      10/01/1990      252.81
Start: 08:31:09.424481
End:    08:31:18.461057
```

If an error occurs it might be handled on

<https://cwatm.iiasa.ac.at/tutorial.html#test-the-python-model-version>

Installation of software 5

CWatM GitHub (not compulsory)

Our software is open source.

We use a public repository to store our software – Github

<https://github.com/iiasa/CWatM>

As CWatM is a living software, we add thing, change things, remove bugs

And we put our changes alive on GitHub

This workshop gives a short introduction into GitHub

Therefore it would be good if you:

- subscribed to a GitHub account
- install GitHub Desktop (<https://desktop.github.com/>)

<https://github.com/>

<https://docs.github.com/en/free-pro-team@latest/github/getting-started-with-c>

<https://www.wikihow.com/Create-an-Account-on-GitHub>

We have a video explaining Github on Sharepoint:

[SO2/workshop_sofia_12_2024/github](https://sharepoint.iiasa.org/Shared%20Documents/SO2/workshop_sofia_12_2024/github)



Problems

Most problems come from different file systems, folder structures

We try to set up everything with relative path.

1. Please make sure that your folders have a similar structure as the SharePoint

The settings file has a part:

[FILE_PATHS]

PathRoot = ../../Sava

../ jumps back to the previous folder

PathOut = \$(PathRoot)/out_drina

./ uses the folder output in the same folder as the settings file or the directory you are in

PathInit = \$(PathRoot)/init

PathMaps = \$(PathRoot)/input

PathMeteo = \$(PathRoot)/meteo/emo-1

3. If this is not working you can use also absolute path (also with white space)

PathRoot = C:/root directory/second.root/cwatm/cwatm_data

4. If you execute CWatM you can also use absolute path

instead

../cwatm_model/cwatmexe/cwatm.exe settings_mekong30min.ini -l

"C:/root directory/second.root/cwatm/cwatm_model/cwatmexe/cwatm.exe" settings_mekong30min.ini -l (mind the " if there are white spaces)

5. Some other errors we address in:

<https://cwatm.iiasa.ac.at/tutorial.html#test-the-python-model-version>